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Prevalence and seasonal dynamics of arthropodan ectoparasites infesting livestock in Muzaffarpur District, Bihar, India

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Abstract- In the current study, an attempt was made to record prevalence and seasonal distribution of arthropodan ectoparasitic infestations on livestock at Muzaffarpur in Bihar. A two years survey was carried out from January, 2023 to December, 2024 in ten villages. A total of 100 (40 cattle, 35 buffaloes and 25 goats) animals were assessed for pathology by the method of visual observation and collection manually. Of all the analysed animals, 62 % were infested with at minimum one species of ectoparasite. The main parasites were ticks (*Rhipicephalus microplus*), lice (*Haematopinus* spp., and mites. Of these, *Rhipicephalus microplus* had the highest frequency (36%) and *Haematopinus* spp. (18%) and *Boophilus annulatus* (8%). The highest infestation rates occurred during the monsoon (July-September, 78%) and the lowest in winter (December-February, 40%). The sites of attachment varied among parasites, with ticks predominantly attached at the dewlap and perineum, lice on the neck and shoulders, and mites mostly at the internal femurs and tail base. The present study reveals that ectoparasitism is still a major health challenge for livestock in Muzaffarpur and the urgent requirement of the integrated parasite management to enhance the productivity of cattle.

Keywords: Ectoparasites, Domestic animals, Ticks, Lice, Mite, Seasonal prevalence

INTRODUCTION

Rhipicephalus microplus is the most related ectoparasite for cattle and buffaloes in tropical and subtropical regions because of its great economic effect.[1] Infestations of lice, primarily due to *Haematopinus* spp., lead to impatience, reduced weight gain and lower milk production.[2] Also, the mites as *Sarcoptes scabiei* causes itch that results in severe itching, skin lesions and secondary bacterial infections.[3] The infestation level and infection pressure of these ectoparasites be contingent primarily on the climatic variables such as temperature, humidity and rainfall, which is why this type of study would be significant seasonally.[4]

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While some studies have described ectoparasite infestations in livestock across India, data from Muzaffarpur district are insufficient. Therefore, the present study was considered to examine the prevalence, species conformation, and seasonal dynamics of ticks, lice, and mites escape livestock in this region.

MATERIALS & METHODS

Study Area

In Bihar, India, in the Muzaffarpur district, between latitudes 26°07' and 26°50' north and longitudes 84°53' and 85°45' east, the present study was conducted. The district receives a subtropical moist climate due to its mid-position within the Gangetic Plain.

Summer is in March through April. It's hot and dry

at 400 degrees Celsius. June to September is the rainy season. The city is very humid and receives lots of precipitation during these months. The relatively cold months of winter, i.e., December to February, having minimum temperatures ranging from 8 to 10 °C, are responsible for seasonal fluctuations in the population of various ectoparasites.[4,5]

Sampling

For ten randomly selected villages from January 2023 to December 2024, monthly surveys were conducted in lieu of various management systems (stall-fed and free-grazing). 45 cattle, 35 buffaloes, and 25 goats were some of the 100 livestock animals studied. Both sexes of animals and different age groups were taken into consideration in order to maintain representative sampling. Farmers were chosen based on their preference and simplicity to use. During the day, clinical checks were performed with special emphasis on sites like the neck, tail base, dewlap, and perineal areas that are at risk from ectoparasites.[6]

Collection and Identification

Ectoparasites were made manually with fine forceps, combing, and skin scrapings from alleged lesions. Specimens collected were preserved in 70% ethanol for laboratory analysis. Identification was done under stereomicroscopes and standard taxonomic keys.[6,7] Ticks were identified to species level, whereas lice and mites were identified to genus level.

Data Analysis

Prevalence (%) was computed with the formula:

Prevalence (%) = Number of infected animals × 100
Total examined

Seasonal spread was compared by categorizing data into four seasons: summer (March- May), monsoon (June-September), post-monsoon (October-November), and winter (December-February). Seasonal prevalence was compared between host species, and descriptive statistics were utilized to approximate differences.

RESULTS

Host-Wise Prevalence

Sixty-two percent of the 100 livestock animals noticed during the study had one or more ectoparasite invasions. Among the diverse host species, cattle presented the maximum frequency, with 27 out of 40 animals (67.5%) infested. Buffaloes were next, with a 60% occurrence of infested animals (21 out of 35). With 14 out of 25 animals (56%) infested, goats had the lowest occurrence. Although the three host groups varied, ectoparasites were present in all species. This implies that invasion is a common problem among the livestock in Muzaffarpur. Given their greater body surface area and closer ties to open grazing, which increases their interaction with ectoparasites, cattle may be more mutual than other animals.

These results highlight the significant problem of ectoparasites in local livestock populations and their possible to affect animal health and efficiency.

Table 1- Host-wise prevalence of ectoparasites

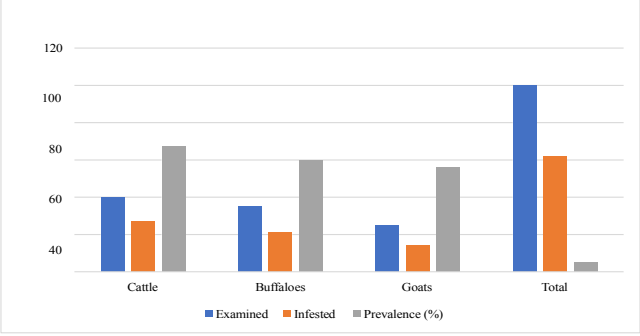
Host	Examined	Infested	Prevalence (%)
Cattle	40	27	67.5
Buffaloes	35	21	60.0
Goats	25	14	56.0
Total	**100**	**62**	**62.0**

On the present research, *Rhipicephalus microplus* was the most prevalent ectoparasite infesting livestock at Muzaffarpur (36%), followed by *Boophilus annulatus* (8%), and *Haematopinus* spp. (18%). Five percent of animals were infested with mixed infestations, which are two or more ectoparasite species (Table 2). The perineal and dewlap area were the main sites for ticks (*R. microplus*), the shoulder and neck regions were the main sites for lice (*Haematopinus* spp.), and the tail base and inner thighs were the main sites for mites (*S. scabiei*).

Table 2- The frequency of various ectoparasite species

Ectoparasite	Prevalence (%)	Preferred Sites
Rhipicephalus microplus	36.0	Dewlap, perineal region
Haematopinus spp.	18.0	Neck, shoulders
Boophilus annulatus	8.0	Inner thighs, tail base

Graph 1 - Host-wise prevalence of ectoparasites



Seasonal variation was superficial in ectoparasite infestations. Post-monsoon (78%) followed by monsoon (67%), summer (58%), and winter (40%) had the highest prevalence, as presented in Figure 1. Though winter atmospheres look less favorable, the high humidity and

good temperature in the monsoon season provide ideal conditions for parasite reproduction and survival.

Livestock ectoparasitism incidence is seasonal, as illustrated in the bar graph. The highest incidence is during the monsoon period (79%), decreases during post-monsoon (68%), further decreases during summer (59%), and is lowest during winter (41%).

DISCUSSION

This study's complete occurrence of 62% is in line with earlier discoveries from Eastern India.[5] Tick populations flourish in moist areas, which is consistent with the majority of *R. microplus*.[8] Monsoon seasonal peaks support earlier findings that ectoparasite dynamics are partial by temperature and humidity.[9] Goats had lower levels of infestation than grazing cattle and buffaloes.

CONCLUSION

The present study highlights the serious problem of ectoparasites in livestock in Muzaffarpur, where R. microplus is the predominant species. Seasonal patterns highlight the need for observation and integrated management, primarily during monsoon months, to minimize parasite-related losses and protect livestock productivity.

REFERENCES

1. **Walker, A. R., Bouattour, A., Camicas, J. L., Estrada-Peña, A., Horak, I. G., Latif, A. A., Pegram, R. G., & Preston, P. M. 2003.** *Ticks of Domestic Animals in Africa: A Guide to Identification of Species*. Bioscience Reports, Edinburgh.
2. **Kumar, R., Singh, P., & Verma, A. 2021.** Prevalence of ectoparasites in cattle and buffaloes in eastern India. *Journal of Parasitic Diseases*, **45(3):** 512–518.
3. **Biswas, S. 2003.** Role of humidity in the survival of ectoparasites. *Indian Journal of Parasitology*, **27(2):** 89–92.
4. **Singh, S. K., Yadav, C. L., & Kumar, S. 2017.** Seasonal prevalence of *Rhipicephalus microplus* in bovines. *Veterinary World*, **10(6):** 688–692.
5. **Kumar, S., Singh, H., & Verma, A. 2021.** Prevalence of ticks in cattle and buffaloes of Eastern India. *Journal of Parasitic Diseases*, **45(3):** 650–657.
6. **Soulsby, E. J. L. 1982.** *Helminths, Arthropods and Protozoa of Domesticated Animals*. 7th ed. Baillière Tindall, London.
7. **Walker, A. R., Bouattour, A., Camicas, J. L., Estrada-Peña, A., Horak, I. G., Latif, A. A., Pegram, R. G., & Preston, P. M. 2003.** Ticks of Domestic Animals in Africa: A Guide to Identification of Species. *Bioscience Reports, Edinburgh*.
8. **Singh, A. P., Sharma, A., & Pathak, M. L. 2017.** Seasonal dynamics of tick infestations in cattle and buffaloes. *Indian Journal of Veterinary Parasitology*, **31(2):** 145–149.
9. **Biswas, S. 2003.** Role of veterinarians in the care and management during harvest of skin in livestock species. *Proc. XXXI Annual Congress of Indian Society for Veterinary Surgery*, IVRI, Izatnagar, 106–110.

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